

1) Given

$$\exists x (P(x) \rightarrow Q(x)) \rightarrow 0$$

$$\forall x P(x) \rightarrow 2$$

$$P(x_0) \rightarrow Q(x_0) \text{ Existential Inst } 0$$

$$P(x_0) \text{ Universal inst } 2$$

$$Q(x_0) \text{ modus ponens } 0, 4$$

$$\text{So } x = x_0 \rightarrow Q(x_0) \text{ Our solution}$$

see its proof writing in Book

Goal

$$\forall x P(x) \rightarrow \exists x Q(x)$$

$$\exists x Q(x)$$

2) we can prove using p.v.q. but since we haven't studied that, we will use proof by Contradiction

A & B are disjoint

$$\neg \exists x [x \in A \wedge (x \in B \wedge x \notin C)]$$

let x be arbitrary

$$x \in A \cap B = x \in A \wedge x \in B$$

$$x \notin C$$

$$A \cap B \subseteq C$$

$$\forall x [x \in (A \cap B) \rightarrow x \in C]$$

$$x \in C$$

Contradiction

Since $x \in B$ & $x \notin C$ & $x \in B \cap C$

but $x \in A$ which is a contradiction

3) $A \subseteq B \cap C$

$$\forall x (x \in A \rightarrow x \in B \cap C) \rightarrow 0$$

$$x \in A \rightarrow 0$$

$$x \in B \cap C$$

let x be arbitrary

Universal inst 0 & 0

A & C are disjoint

$$\neg \exists x (x \in A \wedge x \in C)$$

$$\forall x (x \notin A \vee x \notin C)$$

$$\forall x (x \in A \rightarrow x \notin C)$$

$$x \notin C$$

$$4) A \subseteq P(A)$$

$$P(A) \subseteq P(P(A))$$

$$\forall x (x \in A \rightarrow x \in P(A)) \text{ --- (1)}$$

$$\forall x [x \in P(A) \rightarrow x \in P(P(A))]$$

$$x \in P(A) \text{ --- (2)}$$

$$x \in P(P(A))$$

$$x \subseteq A \text{ --- (3)}$$

$$x \subseteq P(A)$$

$$\forall y (y \in x \rightarrow y \in A) \text{ --- (4)}$$

$$\forall y (y \in x \rightarrow y \in P(A))$$

$$\forall y [y \in x \rightarrow y \subseteq A]$$

$$y \in x \text{ --- (5)}$$

$$y \subseteq A$$

$$y \in A \text{ --- (6) \& (5)}$$

~~$$\forall x (x \in y \rightarrow x \in A)$$~~

$$\forall z (z \in y \rightarrow z \in A)$$

$$x \in y \text{ --- (7)}$$

$$x \in A$$

$$z \in A \text{ --- (8)}$$

Contradiction

$$y \in P(A) \text{ --- (1) \& (6)}$$

$$y \subseteq A \text{ By def}$$

$$\forall x (x \in y \rightarrow x \in A) \text{ --- (9) By def}$$

$$x \in A \text{ Contradiction --- (7) \& (9)}$$

(This can also be proved without contradiction)

$$5) (a) \emptyset \quad (b) P(\emptyset) \text{ as } P(\emptyset) = \{\emptyset\}$$

$$6) (a) \text{ let } x \text{ be arbitrary}$$

$$\forall x [x \neq 1 \rightarrow \exists y (\frac{y+1}{y-2} = x)]$$

$$x \neq 1$$

$$\exists y (\frac{y+1}{y-2} = x)$$

$$\text{let } y = \frac{2x+1}{x-1} \text{ (we found this by solving)}$$

$$\frac{y+1}{y-2} = x$$

$$y+1 = x(y-2)$$

$$y+1 = xy - 2x$$

$$2x - xy + y + 1 = 0$$

$$2x - y(x-1) + 1 = 0$$

$$y(x-1) = 2x+1 \rightarrow y = \frac{2x+1}{x-1}$$

$$\text{Let } \frac{2x+1}{x-1} + 1$$

$$\frac{2x+1}{x-1}$$

-2

$$\rightarrow \frac{2x+1+x-1}{x-1}$$

$$\frac{2x+1-2x+2}{x-1}$$

$$\rightarrow 3x$$

$$\frac{3x}{x-1}$$

$$\rightarrow \frac{3x}{3} = x$$

Since $x \neq 1$, $x-1 \neq 0$.

(b) $\emptyset$

$$\text{Let } y = \frac{2x+1}{x-1}$$

Suppose $x=1$

$$\text{Then } \frac{2(1)+1}{1-1} = \frac{3}{0} = \infty$$

which is not defined (Contradiction)

$$\exists y \left[\frac{y+1}{y-2} = x \rightarrow x \neq 1 \right]$$

$$x \neq 1$$

Contradiction

7) $x \in \mathbb{R}$

Let x be arbitrary

$$x \geq 2$$

$$\text{Let } y = \frac{1}{x-1}$$

$$\forall x \geq 2 \rightarrow \exists y (x+1/y = x)$$

$$\exists y (y+1/y = x)$$

$$y+1/y = x$$

$$\frac{1}{x-1} + 1 \rightarrow \frac{1+x-1}{x-1}$$

$$\frac{1}{x-1}$$

$$\frac{1}{x-1}$$

$$= \frac{x}{x-1}$$

$$\frac{1}{x-1}$$

$$= x$$

Since $x \geq 2$, $x-1 \neq 0$

8) $A \subseteq F \rightarrow A \subseteq U \cap F$
 $A \subseteq U \cap F$
 $\forall x [x \in A \rightarrow x \in U \cap F]$
 let x be arbitrary
 $x \in A$ $\rightarrow$ $x \in U \cap F$
 $\exists x \in F (x \in A)$

9) $A \subseteq F \rightarrow \textcircled{1}$ $\cap F \subseteq A$
 $\forall x [x \in \cap F \rightarrow x \in A]$
 arbitrary x
 $x \in \cap F$ $\rightarrow$ $x \in A$
 $\forall x \in F (x \in A)$
 $\forall x (x \in F \rightarrow x \in A) \textcircled{2}$
 $x \in A$ modus ponens $\textcircled{1} \& \textcircled{2}$

10) $F \neq \emptyset$
 $\forall A \subseteq F (B \subseteq A)$ $B \subseteq \cap F$
 $\forall A (A \subseteq F \rightarrow B \subseteq A) \rightarrow \textcircled{1}$ $\forall x [x \in B \rightarrow x \in \cap F]$
 arbitrary x
 $x \in B$ $\rightarrow$ $x \in \cap F$
 $\neg \exists x \in F (x \in B)$
 arbitrary A
 $A \subseteq F \textcircled{2}$
 $B \subseteq A$ (modus ponens $\textcircled{1} \& \textcircled{2}$) $\rightarrow$ $x \in A$
 $x \in B \& x \in A$

$$11) \phi \in F$$

$$\cap F = \phi$$

$$= \neg \exists x (x \in \cap F)$$

$$= \forall x [\exists A \in F (x \notin A)]$$

let x be arbitrary

$$\exists A \in F (x \notin A)$$

let $A = \phi$, since $\phi \in F$ & ϕ is empty set so $x \notin \phi$.

$$12) F \subseteq G$$

$$UF \subseteq UG$$

$$\forall y [y \in F \rightarrow y \in G] \text{ --- ①}$$

$$\forall x [x \in UF \rightarrow x \in UG]$$

arbitrary x

$$x \in UF$$

$$x \in UG$$

$$\exists A \in F (x \in A) \text{ --- ②}$$

$$\exists B \in G (x \in B)$$

let $A = A_0$.

$A_0 \in F (x \in A_0)$ from ② existential inst

so $A_0 \in G$ Universal Inst ①

so we found a set that belongs to G & x belongs to it.

$$13) F \subseteq G$$

~~$$UF \subseteq UG$$~~

$$\cap G \subseteq \cap F$$

$$\forall x (x \in F \rightarrow x \in G) \text{ --- ①}$$

$$\forall y [y \in \cap G \rightarrow y \in \cap F]$$

let y be arbitrary

$$y \in \cap G$$

$$y \in \cap F$$

$$\forall A \in G (y \in A)$$

$$\forall B \in F (y \in B)$$

$$= \forall A (A \in G \rightarrow y \in A) \text{ --- ②}$$

$$= \forall B (B \in F \rightarrow y \in B)$$

let B be arbitrary

$B \in F$
 $y \in B$
 $B \in G : \text{universal test } \textcircled{1}$
 $y \in B : \text{Universal test } \textcircled{2}$

Acuse Proved

(14)

$$\bigcup_{i \in I} P(A_i) \subseteq P(\bigcup_{i \in I} A_i)$$

$$\forall x [x \in \bigcup_{i \in I} P(A_i) \rightarrow x \in P(\bigcup_{i \in I} A_i)]$$

let x be arbitrary

$$x \in \bigcup_{i \in I} P(A_i)$$

$$x \in P(\bigcup_{i \in I} A_i)$$

$$\exists i \in I (x \in P(A_i))$$

$$x \subseteq \bigcup_{i \in I} A_i$$

$$\exists i \in I (x \subseteq A_i) - \textcircled{2}$$

$$\forall y [y \in x \rightarrow y \in \bigcup_{i \in I} A_i]$$

let y be arbitrary

$$y \in x$$

$$y \in \bigcup_{i \in I} A_i$$

$$\exists i \in I (y \in A_i)$$

$$y \in A_{i_0}$$

let $i = i_0$ for which $x \subseteq A_{i_0}$

($\because$ existential test $\textcircled{2}$)

$$\text{since } y \in x \rightarrow y \in A_{i_0}$$

proved

(15) If $\emptyset$

$$\bigcap_{i \in I} A_i \in \bigcap_{i \in I} P(A_i)$$

let $X = \bigcap_{i \in I} A_i$ we will show $X \in \bigcap_{i \in I} P(A_i)$

$$\forall x [x \in X \rightarrow \forall i \in I (x \in A_i)]$$

$$= \forall i \in I x \in P(A_i)$$

$$\forall x [x \in X \rightarrow \forall i \in I (x \in A_i) \wedge \forall i \in I (x \in A_i) \rightarrow x \in X] - \textcircled{2}$$

let x be arbitrary, let i be arbitrary since $I \neq \emptyset$

Using conjunction from ②

$$\forall x [x \in X \rightarrow x \in A_i]$$

$$\text{So } x \subseteq A_i$$

$$= x \in P(A_i)$$

$$= x \subseteq A_i \quad (\text{Def of } P(A_i))$$

16) $F \subseteq P(B)$

$$U \subseteq B$$

$$\forall x [x \in F \rightarrow x \in P(B)] \text{ --- ①}$$

$$\forall y [y \in U \rightarrow y \in B]$$

Let y be arbitrary

$$y \in U$$

$$y \in B$$

$$\exists A \in F (y \in A) \text{ --- ②}$$

$$A_0 \in F \wedge y \in A_0 \quad \text{--- Existential inst ②}$$

$$A_0 \in P(B)$$

--- Modus ponens & Universal inst ①

$$A_0 \subseteq B$$

inst ①

$$\text{Since } y \in A_0 \text{ \& } A_0 \subseteq B \rightarrow y \in B.$$

17) $F \neq \emptyset, G \neq \emptyset$

$$\forall x [x \in F \rightarrow \forall y (y \in G \rightarrow x \subseteq y)] \text{ --- ①} \quad U \subseteq G \cap G$$

$$\forall z [z \in U \rightarrow z \in G]$$

Let x be arbitrary

$$x \in U$$

$$x \in G \cap G$$

$$\forall B \in G (x \subseteq B)$$

Let B be arbitrary

$$x \subseteq B$$

$$\exists A \in F (x \in A) \text{ --- ②}$$

$$\text{Let } A_0 \in F \text{ and } x \in A_0 \quad \text{--- from ②}$$

$$B \in G$$

$$\text{Since } A_0 \in F \text{ so } \forall y (y \in G \rightarrow A_0 \subseteq y) \text{ --- modus ponens \& instantiation ①}$$

Since $B \in G$, so $A \cup B \subseteq G$

$\therefore$ Universal test, Modulus (3)

Since $2 \in A \cup B$ & $A \cup B \subseteq G$

$\therefore 2 \in G$

18) (a) $a \mid b, a \mid c$

$a \mid (b+c)$

$\exists n (na = b)$

$\exists o (ao = b+c)$

$\exists m (ma = c)$

We can use 'ad m' for installation but
for easiness let's use n & m

$na = b, ma = c$

Adding we get $na + ma = b + c$

$(n+m)a = b+c$

So let $o = n+m \rightarrow ao = b+c$ so $a \mid b+c$

(b) $ac \mid bc$

$a \mid b$

$c \neq 0$

$\exists m (am = b)$

$\exists n (nac = bc)$

$noa \neq b \neq$ ($\therefore$ let $n = no$)

$noa = b$

So let $m = no$ ad $noa = b$ so $a \mid b$

19) ^{prove} $\forall x \forall y \exists z (x + z = y)$

let x & y be arbitrary R

$$\exists z (x + z = y)$$

$$\downarrow$$

$$x + z = y$$

$$z = \frac{y - x}{2}$$

let $z = \frac{y - x}{2}$

$$x + \frac{y - x}{2} = \frac{y - y + x}{2}$$

$$\frac{2x + y - x}{2} = \frac{2y - y - x}{2}$$

$$\frac{y}{2} = \frac{y - x}{2}$$

$$y - x = y - x \text{ which satisfies}$$

(b) No $\forall x \exists y$ $y - x$ may or may not be an integer

25) $\forall x \exists y \forall z [yz = (xz)^2 - (x^2 + z^2)]$
 where x is arbitrary

suppose $y = 2x$

$$\text{then } yz = 2xz = x^2 + z^2 + 2xz - x^2 + z^2$$

$$2xz = 2xz$$

Proved. Note that to define y , we only used x , not z because z is defined after y .

18. In this problem all variables range over $\mathbb{Z}$, the set of all integers.
- Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
 - Prove that if $ac \mid bc$ and $c \neq 0$, then $a \mid b$.
19. (a) Prove that for all real numbers x and y there is a real number z such that $x + z = y - z$.
- Would the statement in part (a) be correct if "real number" were changed to "integer"? Justify your answer.
- *20. Consider the following theorem:

Theorem. For every real number x , $x^2 \geq 0$.

What's wrong with the following proof of the theorem?

Proof. Suppose not. Then for every real number x , $x^2 < 0$. In particular, plugging in $x = 3$ we would get $9 < 0$, which is clearly false. This contradiction shows that for every number x , $x^2 \geq 0$. $\square$

21. Consider the following incorrect theorem:

Incorrect Theorem. If $\forall x \in A (x \neq 0)$ and $A \subseteq B$ then $\forall x \in B (x \neq 0)$.

- What's wrong with the following proof of the theorem?

Proof. Suppose that $\forall x \in A (x \neq 0)$ and $A \subseteq B$. Let x be an arbitrary element of A . Since $\forall x \in A (x \neq 0)$, we can conclude that $x \neq 0$. Also, since $A \subseteq B$, $x \in B$. Since $x \in B$, $x \neq 0$, and x was arbitrary, we can conclude that $\forall x \in B (x \neq 0)$. $\square$

- Find a counterexample to the theorem. In other words, find an example of sets A and B for which the hypotheses of the theorem are true but the conclusion is false. $A = \{1, 2\}$ $B = \{1, 2, 0\}$

- *22. Consider the following incorrect theorem:

Incorrect Theorem. $\exists x \in \mathbb{R} \forall y \in \mathbb{R} (xy^2 = y - x)$.

What's wrong with the following proof of the theorem?

Proof. Let $x = y/(y^2 + 1)$. Then

$$y - x = y - \frac{y}{y^2 + 1} = \frac{y^3}{y^2 + 1} = \frac{y}{y^2 + 1} \cdot y^2 = xy^2. \quad \square$$

23. Consider the following incorrect theorem:

Incorrect Theorem. Suppose $\mathcal{F}$ and $\mathcal{G}$ are families of sets. If $\bigcup \mathcal{F}$ and $\bigcup \mathcal{G}$ are disjoint, then so are $\mathcal{F}$ and $\mathcal{G}$.

- (a) What's wrong with the following proof of the theorem?

Proof. Suppose $\bigcup \mathcal{F}$ and $\bigcup \mathcal{G}$ are disjoint. Suppose $\mathcal{F}$ and $\mathcal{G}$ are not disjoint. Then we can choose some set A such that $A \in \mathcal{F}$ and $A \in \mathcal{G}$. Since $A \in \mathcal{F}$, by exercise 8, $A \subseteq \bigcup \mathcal{F}$, so every element of A is in $\bigcup \mathcal{F}$. Similarly, since $A \in \mathcal{G}$, every element of A is in $\bigcup \mathcal{G}$. But then every element of A is in both $\bigcup \mathcal{F}$ and $\bigcup \mathcal{G}$, and this is impossible since $\bigcup \mathcal{F}$ and $\bigcup \mathcal{G}$ are disjoint. Thus, we have reached a contradiction, so $\mathcal{F}$ and $\mathcal{G}$ must be disjoint.

- (b) Find a counterexample to the theorem.

24. Consider the following putative theorem:

Theorem? For all real numbers x and y , $x^2 + xy - 2y^2 = 0$.

- (a) What's wrong with the following proof of the theorem?

Proof. Let x and y be equal to some arbitrary real number r . Then

$$x^2 + xy - 2y^2 = r^2 + r \cdot r - 2r^2 = 0.$$

Since x and y were both arbitrary, this shows that for all real numbers x and y , $x^2 + xy - 2y^2 = 0$.

- (b) Is the theorem correct? Justify your answer with either a proof or a counterexample.

- *25. Prove that for every real number x there is a real number y such that for every real number z , $yz = (x + z)^2 - (x^2 + z^2)$.

26. (a) Comparing the various rules for dealing with quantifiers in proofs, you should see a similarity between the rules for goals of the form $\forall x P(x)$ and givens of the form $\exists x P(x)$. What is this similarity? What about the rules for goals of the form $\exists x P(x)$ and givens of the form $\forall x P(x)$?

- (b) Can you think of a reason why these similarities might be expected?

(Hint: Think about how proof by contradiction works when the goal starts with a quantifier.)

3.4. Proofs Involving Conjunctions and Biconditionals

The method for proving a goal of the form $P \wedge Q$ is very simple:

To prove a goal of the form $P \wedge Q$:

Prove P and Q separately.

We are making assumption that A is not empty. We can only assume that $A \in \mathcal{F} \wedge A \in \mathcal{G}$.

$A \in \mathcal{F} \wedge A \in \mathcal{G}$ can also be assumed since it is a proven theorem that $A \in \mathcal{F} \wedge A \in \mathcal{G}$ cannot be assumed.

We are making assumption that $x = y = r$. If x and y may be different.

$$x^2 + xy - 2y^2 = r^2 + r \cdot r - 2r^2 = 0.$$

Since x and y were both arbitrary, this shows that for all real numbers x and y , $x^2 + xy - 2y^2 = 0$. $x=3, y=2$ is not proven.

$\mathcal{F} = \{\{2, 13\}, \{1, 2\}\}$
 $\mathcal{G} = \{\{2, 13\}, \{1, 2\}\}$
 $\bigcup \mathcal{F} = \{1, 2, 13\}$
 $\bigcup \mathcal{G} = \{1, 2, 13\}$
 $\bigcup \mathcal{F} \cap \bigcup \mathcal{G} = \{1, 2, 13\} \neq \emptyset$

Goal of form $\forall x P(x)$ can be converted to $\exists x \neg P(x)$ by contradiction & vice versa $\rightarrow \exists x P(x)$ can be converted to $\forall x \neg P(x)$

In both, we introduce a new variable x into proof. but doesn't give value of x . In both cases, we put a particular value of x in $P(x)$.